

Emmanuelle Coutu-Nadeau, B.Eng

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Education

Master of Science, Psychology (GPA: 4.2 / 4.3)	University of Montreal	Sep 2024 – Present
Bachelor of Software Engineering (GPA: 3.7 / 4.0)	McGill University	Sep 2018 – May 2022
Bachelor of Science, Psychology, Honours (GPA: 3.8 / 4.0)	McGill University	Sep 2015 – May 2018

Awards & Distinctions

Neuro-AI UNIQUE Excellence Scholarship	\$10,000	Sep 2025
MES Bourse de perseverance TI	\$3,000	Mar 2022
Airbus Scholarship in Engineering	\$1,500	May 2021
First-Class Honours	-	May 2018

Research Experience

Master Student Researcher – Neurosciences of Early Development Lab (CHU Ste-Justine)	Sep 2024 – Present
• M.Sc. Thesis: Developed a ML approach to identify transdiagnostic behavioral profiles and associated EEG signatures in children and adolescents with neurodevelopmental conditions. • Conducted EEG recordings with youth presenting ASD and other neurodevelopmental conditions. • Database Standardization: Led a weekly initiative to modernize the lab's database architecture; evaluated database solutions and established unified standards for code sharing.	
Research Assistant – Co-venture Lab (CHU Ste-Justine)	Nov 2017 – Jul 2018
• Assisted with data entry for a study on substance use and adolescent cognitive development.	
Honours Trainee – Martel Pain Lab (McGill University, MUHC)	Oct 2016 – May 2019
• B.Sc. Honours Thesis: Association between perceived injustice and prescription opioid misuse. • Conducted structured clinical interviews, ecological momentary assessments and saliva sampling.	

Peer-reviewed Journal Articles

1. **Coutu-Nadeau, E.**, Arutiunian, V., Martin, C.-O., et al. (under review). Behavioral Clustering and Resting-State EEG Signatures of Transdiagnostic Neurodevelopmental Profiles. <https://doi.org/10.21203/rs.3.rs-10692457/v1>
2. Arutiunian, V., Davoudi, S., Knoth, S.I., **Coutu-Nadeau, E.**, et al. (under review). Auditory Sustained Brain Activity Links Genomic Variation to Language in Neurodevelopmental Disorders.
3. Verner M, Sirois A, **Coutu-Nadeau E**, et al. The association between perceived injustice and opioid craving in patients with chronic pain. J Addict Med. 2022 Jul 21. <https://doi.org/10.1097/adm.0000000000001022>

Professional Experience

Product Owner – CareSimple	Oct 2022 – Sep 2025
• Translated client needs into technical specs for a digital health platform. • Developed data-driven reporting tools using Python and SQL.	
Software Engineer – CareSimple	May 2020 – Oct 2022
• Developed features on a web-based portal for clinicians to perform remote patient monitoring.	

Clinical Assessment Training

Children Memory Scale (CMS)	May 2026
California Verbal Learning Test for Children (CVLT-C)	May 2026
Weschler Intelligence Scale for Children (WISC-V)	Apr 2025
Weschler Adult Intelligence Scale (WAIS-IV)	Apr 2025
Bear-Dragon Inhibition Test	Mar 2025
Identifying People Vulnerable to Suicide and Implementing Protective Measures – AQPS	Sep 2024
Bayley Scales of Infant and Toddler development-IV (Bayley (IV)	Aug 2024
Structured Clinical Interview for DSM-5 (SCID-5) – Substance Withdrawal	Aug 2017

Clinically Relevant Experience

Patient & Bedside Support Volunteer – CHU Ste-Justine	October 2026 –
Starting October 2026.	
Crisis Responder – Kids Help Phone	Jul 2026 – Present
Developed skills in text-based empathetic communication, collaborative safety planning, and real-time risk management.	
Pediatric Oncology, Cognitive Evaluator – CHU Ste-Justine	Sep 2025 – Present
Administered standardized cognitive assessments to children and adolescents in oncology remission and practiced active listening to address parental concerns.	
Mentor – Technovation Montreal	Jan 2024 – May 2026
- Mentored five groups of girls (ages 8–18) from January to May, guiding them through mobile app and business plan development to address a UN Sustainable Development Goal. - Facilitated group dynamics and mediated interpersonal conflict within teams, fostering self-confidence, while providing ongoing support to both youth and parents throughout the program.	
Clinical Research Assistant (Substance use) – Montreal General Hospital (MUHC)	Oct 2016 – May 2019
Conducted structured clinical interviews with adult patients experiencing chronic pain and prescription opioid use disorders.	

Community Engagement

La Maison du Père – Board Member	Sep 2026 – Present
Collaborated with executive leadership to review annual budgets, fundraising goals, and long-term community impact strategies for a non-profit organization serving men experiencing homelessness.	
UNIQUE Centre – Student Affairs Committee	Oct 2025 – Present
- Organized events and curated research highlights to support students at the intersection of Neuroscience and AI across Quebec. - Participated in a panel discussion on the transition from industry to academia.	
MAC Foundation – Event Ambassador	Jan 2025 – Apr 2026
Promoted for “Les Printemps du MAC” 2025 and 2026 editions, an initiative supporting Montreal’s Museum of Contemporary Arts through fundraising events.	
Pensionnat du Saint-Nom-de-Marie Foundation	Jan 2023 – Apr 2026
- Led and mentored a group of high school aged youth during a two-day bootcamp, facilitating structured leadership workshops and team-building activities. - Spoke at Career Day events and helped organize annual panels with school alumni.	
Promoting Opportunities for Women in Engineering (POWE)	Feb 2019
Led lab visits at The Neuro, introducing high-school girls to neuroscience and biomedical engineering.	
Meals On Wheels “Santropol Roulant”	Nov 2016 – Jun 2022
Supported community food security through meal preparation, home delivery for homebound individuals, and urban agriculture initiatives.	

Technical Skills

Languages: Fluent in French and English.

Programming & Data analysis: Python, Java, C, MATLAB, EEGLab, React, TypeScript, R, SQL, SPSS, AI/ML.

Certifications: PSPO I, HIPAA, GDPR, Cybersecurity training, Leading Change in Health Informatics.

Personal Accomplishments

Running: Completed four half marathons (21km; Banff 2015, Montreal 2016, Toronto 2018, Montreal 2022)

Hackathons: Participated in McHacks 6 (2019)

Conference Presentations & Abstracts

Coutu-Nadeau E, Arutiunian V, Martin CO, Fontaine V, Bélanger AM, Jizi K, Knoth IS, Hosein A, Davoudi S, Elsabbagh M, Forgeot D'Arc B, Jacquemont S, Lippé S. Behavioral Clustering and Resting-state EEG signatures of Neurodevelopmental Profiles Using Machine Learning. Society for Psychophysiological Research; 2026 Oct ([upcoming](#)); Halifax, NS, Canada.

Coutu-Nadeau E, Arutiunian V, Martin CO, Fontaine V, Bélanger AM, Jizi K, Knoth IS, Hosein A, Davoudi S, Elsabbagh M, Forgeot D'Arc B, Jacquemont S, Lippé S. Behavioral Clustering and Resting-state EEG signatures of Neurodevelopmental Profiles Using Machine Learning. Child Neurology Society; 2026 Oct ([upcoming](#)); Montreal, QC, Canada.

Coutu-Nadeau E, Arutiunian V, Martin CO, Fontaine V, Bélanger AM, Jizi K, Knoth IS, Hosein A, Davoudi S, Elsabbagh M, Forgeot D'Arc B, Jacquemont S, Lippé S. Behavioral Clustering and Resting-state EEG signatures of Neurodevelopmental Profiles Using Machine Learning. 93e Congrès de l'Acfas (TACC Trainee Research Day on Autism and Neurodivergence) ; 2026 May 12; Trois-Rivières, QC, Canada.

Coutu-Nadeau E, Arutiunian V, Martin CO, Fontaine V, Bélanger AM, Jizi K, Knoth IS, Hosein A, Davoudi S, O'Reilly C, Rouleau G, Ernst C, Lucas-Tardif C, Scorah J, Couture M, Abadie P, Joober R, Forgeot d'Arc B, Jacquemont S, Elsabbagh M, Lippé S. Resting-state EEG signatures of ASD/ADHD: toward identifying behavioral subgroups. Society for Psychophysiological Research Annual Meeting; 2025 Oct; Montreal, QC, Canada.

Kennedy G, Baran-Goldwax M, Pomerleau R, **Coutu-Nadeau E**, Knoth IS, Lippé S. Hormonal modulation of brain dynamics in menstruating adolescents and young adults: EEG insights across the menstrual cycle. Society for Psychophysiological Research Annual Meeting; 2025 Oct; Montreal, QC, Canada.

Kennedy G, Baran-Goldwax M, **Coutu-Nadeau E**, Pomerleau R, Knoth IS, Lippé S. Feeling everything: estrogen, cortisol, and sensory brain dynamics across the menstrual cycle. International Society of Psychoneuroendocrinology Annual Conference; 2025 Sep; New Orleans, LA, USA.

Herrero Babiloni A, Frimerman L, Fournier C, **Coutu-Nadeau E**, Ross A, Lavigne GJ, Martel MO. Sleep quality and prescription opioid misuse: the potential mediating role of pain intensity, negative affect, and opioid craving. *Sleep Med.* 2019 Dec;64(Suppl 1):S152–S153.

Frimerman L, Fournier C, **Coutu-Nadeau E**, Ross A, Martel MO. Determinants of prescription opioid craving and opioid-related hedonic feelings among patients with chronic pain prescribed opioid therapy. *J Pain.* 2019 Mar;20(4 Suppl): S55–S56.

Herrero Babiloni A, Frimerman L, Fournier C, **Coutu-Nadeau E**, Ross A, Lavigne GJ, Martel MO. Sleep quality and opioid craving among patients with chronic pain and prescribed opioid therapy: explorations of potential mediators. 11th Congress of the European Pain Federation; 2019 Sep; Valencia, Spain.

Herrero Babiloni A, Frimerman L, Fournier C, **Coutu-Nadeau E**, Ross A, Lavigne GJ, Martel MO. Sleep quality and prescription opioid misuse: the potential mediating role of pain intensity, negative affect, and opioid craving. World Sleep Congress; 2019 Sep; Vancouver, BC, Canada.

Frimerman L, Fournier C, **Coutu-Nadeau E**, Ross A, Martel MO. Determinants of prescription opioid craving and opioid-related hedonic feelings among patients with chronic pain prescribed opioid therapy. 23rd Annual McGill Pain Day; 2019 Jan; Montreal, QC, Canada.

Frimerman L, Fournier C, **Coutu-Nadeau E**, Ross A, Martel MO. Determinants of prescription opioid craving and opioid-related hedonic feelings among patients with chronic pain prescribed opioid therapy. 38th Annual Scientific Meeting of the American Pain Society; 2019 Apr; Milwaukee, WI, USA.

Ouellet N, Fournier C, Frimerman L, **Coutu-Nadeau E**, Martel MO. Factors associated with opioid craving among patients with chronic pain prescribed long-term opioid therapy. 14th Annual Research Day of McGill's Faculty of Dentistry; 2019 Apr; Montreal, QC, Canada.